

Print Estimator - Logic Checklist (for NKK sir)

Tick correct / wrong. For every wrong one, write the correct formula in the last column. Live app: andreal print estimator (GitHub Pages).
Updated 2026-06-18.

A. Paper / sheet calculation

#	Column	Formula the app uses	✓ / ✗	Sir's correction
A1	Open size	portrait -> 2xwidth ; landscape -> 2xheight		
A2	Open size with bleed	open size + bleed both sides		
A3	Form (pp)	largest standard fold <= 4xups, from {4,8,12,16,24,32}		
A4	Forms / book	ceil(text pages / form)		
A5	Gang / sheet	floor(4xups / form), capped to forms/book		
A6	Layouts	ceil(forms / gang)		
A7	Good sheets	layouts x copies		
A8	Running waste	formes x copies x running% (formes = layouts x sides)		
A9	Make-ready	layouts x make-ready sheets/forme		
A10	TOTAL sheets	good + running + make-ready		
A11	Reams	total / 500		
A12	Weight / sheet	sheet m2 x gsm / 1000		
A13	Waste %	100 - (form pp x spread area) / (4 x sheet area)		

B. Cut sheet (on press) + parent buying

#	Column	Formula the app uses	✓ / ✗	Sir's correction
B1	Cut sheet (on press)	required imaged size (cols x cell + gripper/side-lay/trim), capped to press		
B2	Cuts per parent	floor(parent / cut), best of both orientations		
B3	Parents to buy	ceil(cut sheets / cuts-per-parent)		
B4	Paper cost & weight	on the PARENT sheet		
B5	When to cut	cut only if a cheaper press needs it; else print whole		

C. Machine selection

#	Column	Formula the app uses	✓ / ✗	Sir's correction
C1	Auto press	cheapest total route (paper+print+plates), whole or cut		
C2	Digital	excluded from Auto (click-rate pending); manual only		
C3	Band	big if max >= 34x22 in, else small		

D. Pricing

#	Column	Formula the app uses	✓ / ✗	Sir's correction
D1	Paper	parent sheets x Rs/sheet		
D2	Printing	per forme: ceil(total/layouts) sheets x lot / per-1000 tier		
D3	Plates	(cf+cb) x layouts x plate rate (digital = 0)		
D4	Coating	rate/100 sq.in x sheets, min lot		
D5	Embellishment (x3)	rate/100 sq.in x cover sheets, min lot		
D6	Binding	Rs/book x copies, min		
D7	Finishing make-ready	(ops) x 10-50 sheets x parent rate		
D8	Packing	ceil(kg / 15) x Rs35		
D9	Freight	Rs8/kg (min Rs300)		

D10	Overhead	12% of (paper+print+plates+coating+binding+packing+freight)		
D11	Margin	28% of production		
D12	GST	18% (CGST 9 + SGST 9 / IGST 18)		

E. Binding / spine

#	Column	Formula the app uses	✓ / ✗	Sir's correction
E1	Spine thickness	(pages / 2) x gsm x bulk / 1000 mm		
E2	<= 2.5 mm	recommend Saddle stitch		
E3	> 3 mm	warn -> Section sewn / Perfect		
E4	> 4.5 mm	block Saddle (Force to override)		
E5	Binding rates	saddle 3.5 / side 4 / section-sewn 4.32 / sewn+perfect 6.75 / perfect 9 / hardcase 95		

F. Still pending sir's data

#	Item	What I need from sir
F1	Wastage % schedule	which quantity -> 15% vs 25% vs 5%
F2	Grain direction	the grain-compatibility rule
F3	24pp / 32pp forms	hand-drawn fold patterns (8/12/16 already match)
F4	Digital Rs/click card	the click-rate table per digital press
F5	Printing/impression tiers	confirm the rate tables

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